

Measuring the Shadow of Intervention

A two-stage machine learning pipeline for latent geopolitical quantities

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The problem

Theory in international relations predicts that *anticipated* third-party military intervention shapes whether civil wars start. If a rebel group expects powerful states to back the government, fighting looks less attractive; if outside support for the opposition is likely, it looks more attractive. The quantity that matters—the **shadow of intervention**—is an expectation about the international environment, not a realized event.

No one has measured it. Existing approaches either use a static proxy (the US security hierarchy) or estimate a structural game restricted to the five permanent Security Council members. But P5 states account for fewer than half of all military interventions; neighbors, regional rivals, and coethnics make up the rest.

This project constructs a measure of expected intervention that covers all ~190 potential interveners, varies annually with shifting alliances and capabilities, and distinguishes government-biased from opposition-biased intervention. The measure is then used to test whether the shadow actually deters and emboldens.

Pipeline architecture

The pipeline has two stages, bridged by a game-theoretic equilibrium computation.

Raw data (COW, V-Dem, UNGA, trade, alliances, ...)

Multiple imputation (miceforest, 5 CY × 5 UD = 25 datasets)

105 directed-dyad-year features

STAGE 1: Super-learner ensemble

9 classifiers (RF, 2×HGB, ridge, elastic-net,
lasso, multinomial logit, 2×MLP) → 10-fold CV
→ NNLS stacking on multinomial log-loss

Spatial lags ← Nash fixed-point iteration
(predictions feed back as inputs until
self-consistent)

Dyad-level $P(\text{intervention} \mid \text{direction})$

Aggregate to country-year shadow:
 $E_G = \Sigma P(\text{gov-biased}), \quad E_O = \Sigma P(\text{opp-biased})$

STAGE 2: Onset logit

Does the shadow predict civil war onset?
T×P bootstrap (25 draws × 200 cluster reps)
propagates measurement uncertainty into SEs

Why the fixed point?

Interveners don't act in isolation. When one state intervenes on the government side, rivals face pressure to back the opposition. The classifier uses spatial lags—summaries of other states' behavior—as features. But out of sample, those lags should reflect the model's own predictions, not historical observations. The **Nash fixed-point iteration** requires predictions to be self-consistent inputs to themselves:

$$\sigma^*(B, A, t) = \hat{M}(X(B, A, t, \sigma^*)) \quad \forall (B, A, t)$$

This is both a measurement correction (eliminating a deterministic discrepancy between lag inputs and probability outputs) and a game-theoretic equilibrium condition. Convergence is rapid—two to three passes—across all 25 imputation datasets.

Data architecture

The unit of analysis for the classifier is a **directed dyad-year**: an ordered pair (potential intervener B , potential host A , year t). For each of ~ 190 states in the international system across 69 years (1946–2014), we ask: would B intervene militarily in A ’s civil conflict, and if so, on which side?

Getting there requires assembling, aligning, and imputing data from a dozen sources at three levels of analysis.

Three levels, one pipeline

Country-year features describe each state in isolation: material capabilities (GDP, military personnel, iron/steel production from COW National Material Capabilities v6), regime type (Polity II, V-Dem liberal democracy index, instability indicators), ethnic fractionalization, oil wealth, terrain, and foreign policy positioning (UNGA ideal points). These come from separate datasets with different country coding schemes, temporal coverage, and missingness patterns.

Undirected-dyad features describe relationships between pairs of states: alliance commitments, bilateral trade, geographic distance, contiguity, colonial history, shared IGO membership, territorial disputes, and peace quality. Each source uses its own dyad identifier format and covers a different time span.

Directed-dyad features assign roles: state A is the potential civil war host, state B the potential intervener. This doubles many features (GDP of A vs GDP of B , regime type of A vs regime type of B) and adds asymmetric quantities like the capability ratio B/A . The final feature matrix has **105 columns** per directed-dyad-year observation.

Staged imputation

Missing data is pervasive—early Cold War GDP figures, pre-1965 ideal points, incomplete alliance records. A complete-case analysis would discard most of the sample.

The imputation uses **miceforest** (MICE with random forests), but the staging matters:

1. **Country-year first.** GDP, regime type, and other state-level variables are imputed at the country-year level, producing 5 complete CY panels. This ensures that a country’s GDP is the same value in every dyad involving that country—imputing at the dyad level would break this consistency.
2. **Undirected-dyad second.** Bilateral variables (trade, alliance similarity, ideal-point distance) are imputed conditional on the already-imputed CY variables, producing 5 UD draws per CY draw = 25 complete datasets.

3. **Directed expansion last.** The undirected dyad (A, B) is expanded into two directed rows $(A \rightarrow B)$ and $(B \rightarrow A)$, with features assigned by role. No imputation here—just reshaping.

The result is 25 complete directed-dyad-year datasets, each around $25,700 \text{ rows} \times 105 \text{ features}$, stored as parquet files ($\sim 92 \text{ MB}$ each for the spatial weight matrices, smaller for the feature tables).

Spatial weight matrices

Ten spatial lag features capture the strategic context: what are other potential interveners doing in the same host country? These are weighted averages across all states $C \neq B$ of their intervention behavior toward A , using a polity-similarity weight matrix (states with similar regime types get higher weight, reflecting ideological alignment in intervention decisions).

The weights are constructed separately for each of the 25 imputation datasets, since the polity scores feeding the similarity calculation are themselves imputed. This produces 25 distinct spatial weight matrices—each $\sim 92 \text{ MB}$ —that propagate imputation uncertainty through the spatial structure.

Stage 1 results

The ensemble achieves a **proportional reduction in log-loss (PRL) of 47.3%** and an AUC of 0.969 on out-of-fold predictions. For a three-class rare-event problem where interventions account for $\sim 1\%$ of observations, this is substantial.

Where does the performance come from? Decomposing the log-loss reduction by true class reveals that **94% of the PRL comes from the 1% of observations where intervention actually occurred**. Only 6% comes from improved calibration of non-events. The classifier earns its keep by finding needles, not by calibrating hay—which matters, because sending all intervention probabilities to zero is exactly what a standard onset model without shadow variables already does.

Ensemble weights (averaged across 25 imputation draws):

Classifier	Weight
Deep MLP (100, 50)	44.5%
Random forest	38.6%
Histogram-based GBM (low LR)	11.1%
Histogram-based GBM	3.7%
All others	<2.1%

The multinomial logit—which corresponds to the softmax best-response function assumed by structural game-theoretic models—earns about 0.4% of the weight. The data prefer nonparametric alternatives.

The shadow measure

Aggregating dyad-level predictions to the country-year level produces two time-varying measures: expected government-biased intervention (E^G) and expected opposition-biased intervention (E^O).

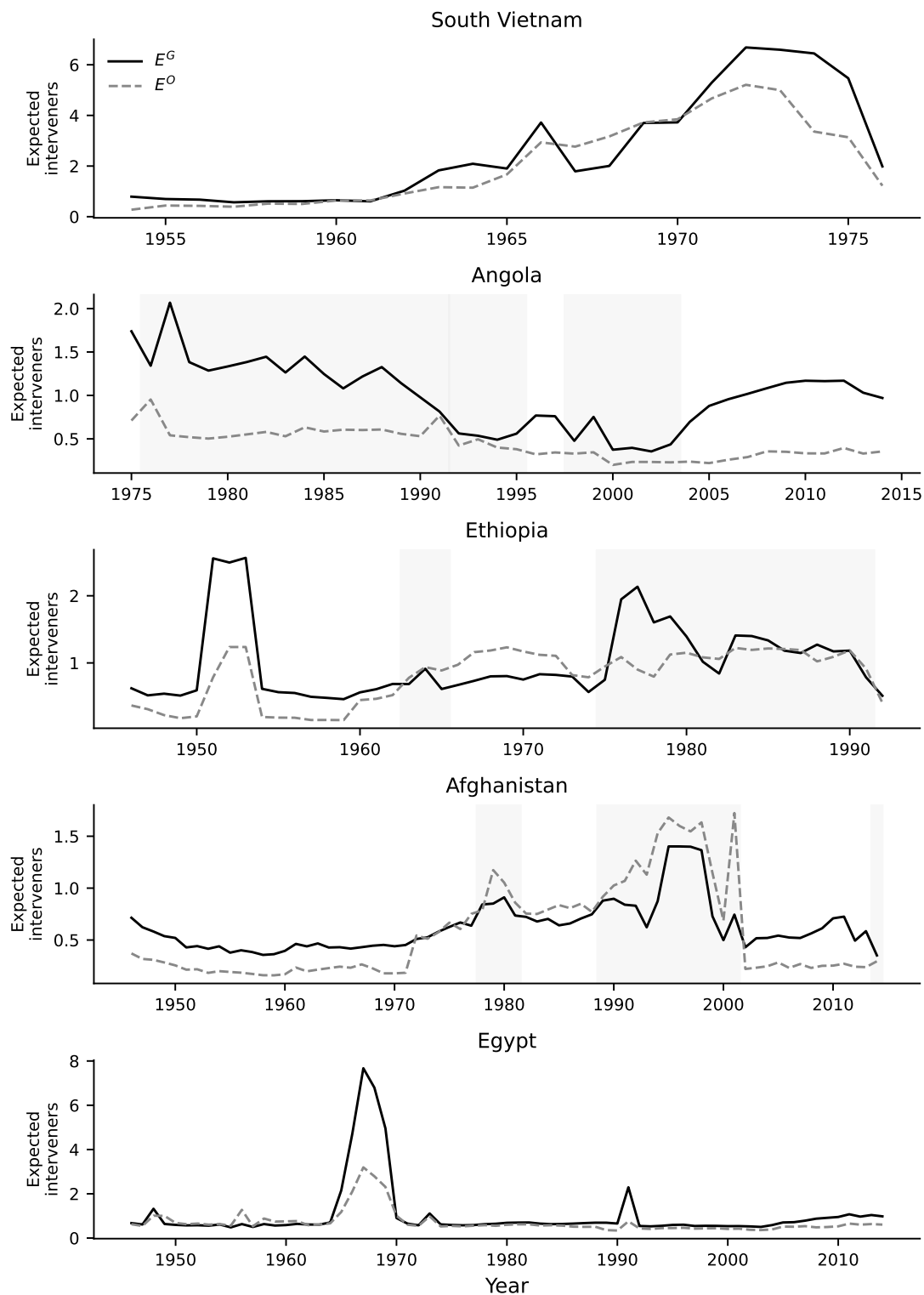


Figure 1: Shadow measure time series for five countries with well-documented intervention environments.

South Vietnam shows the cleanest signal: both measures ramp through the 1960s to peak intervention expectations during the war’s height. Afghanistan captures the regime-switching dynamic of the late 1970s. Egypt—which experiences no civil war in our data—provides a pure out-of-sample check: the shadow tracks the Six Day War, the Soviet expulsion, and the Gulf War purely from counterfactual predictions.

Stage 2: Does the shadow predict onset?

The onset logit asks whether expected intervention enters the civil war calculus as theory predicts.

	Coefficient	T×P SE	95% CI
E^G (gov-biased)	−1.62	1.37	[−4.63, +0.69]
E^O (opp-biased)	+0.64	1.10	[−1.94, +2.59]

The signs are as predicted: government-biased intervention expectations **deter** onset; opposition-biased expectations **embolden** it. The confidence intervals are wide—by design.

Honest uncertainty

The T×P bootstrap propagates measurement-stage uncertainty (variation across the 25 imputation draws) into the standard errors. This inflates SEs by a factor of $3.5\times$ (E^G) and $2.9\times$ (E^O) relative to naive maximum-likelihood estimates. Variance decomposition shows **92% of total uncertainty in E^G is measurement-stage** (88% for E^O), not sampling noise. In other words: the dominant source of uncertainty isn’t “do we have enough civil wars?”—it’s “how precisely can we measure the shadow?”

This is the contribution. Wide confidence intervals from honest uncertainty propagation are more informative than narrow intervals that pretend a learned proxy is observed data.

Technical stack

- **Language:** Python 3.14
- **ML:** scikit-learn (ensemble, logistic, MLP), LightGBM (HGB)
- **Imputation:** miceforest (multiple imputation by chained equations with random forests)
- **Data:** pandas, pyarrow/parquet throughout
- **Paper:** LaTeX
- **Pipeline:** Jupyter notebooks (01–09), reproducible via nbconvert

Code

The full pipeline lives in nine notebooks:

Notebook	What it does
01	Country-year panel construction + multiple imputation
02	Directed-dyad expansion + bilateral features
03	Intervention coding (Regan 1944–1999 + post-1999 extension)
04	Spatial weight matrices (25 datasets × polity-similarity weights)
05	Stage 1 training (super-learner, 10-fold CV, Nash fixed-point)
06	Stage 1 predictions (universal application + aggregation)
07	Stage 2 onset analysis (logit, specifications, T×P bootstrap)
09	Figure generation

Source code for data loading, feature engineering, and model utilities is in `src/shadow/`.